SKF 6205-2RS Deep Groove Ball Bearing Maintenance and Engineering Manual

Section 1: Geometric Specifications and Internal Clearance

The SKF 6205-2RS single-row deep groove ball bearing has a bore diameter of 25.0 mm, an outside diameter of 52.0 mm, and a width of 15.0 mm.

Internal geometry: N = 9 rolling elements (balls), ball diameter d = 7.94 mm, pitch diameter D = 39.04 mm, and nominal contact angle phi = 0.0 degrees.

Standard radial internal clearance (CN) before mounting is 5 to 20 micrometers, while C3 radial internal clearance before mounting is 13 to 28 micrometers.

At a shaft speed of 1797 RPM (29.95 Hz), the theoretical defect frequencies are: FTF = 11.93 Hz, BPFO = 107.36 Hz, BPFI = 162.19 Hz, and BSF = 70.58 Hz.

Section 2: Lubrication Intervals, Grease Selection, and Mounting Fits

For continuous industrial operation at 1500 to 1800 RPM under moderate radial load and bearing housing temperatures below 70 degrees C, the recommended relubrication interval for open/shielded 6205 bearings is 4500 operating hours using lithium-complex ISO VG 100 mineral grease (fill 30 to 35 percent of free internal space).

For sealed 6205-2RS bearings, inspect seals every 2000 hours for grease purging or contaminant ingress. Shaft fit tolerance for rotating inner ring loads should be k5 or m5, and housing bore tolerance for stationary outer ring loads should be H7 or J7.

Section 3: Bearing Fault Progression Stages and Maintenance Actions
Stage 1 (Subsurface micro-spalling): ultrasonic/spike energy in 20 to 60 kHz band; normal time-domain RMS and kurtosis near 3.0. Action: continue normal monitoring.
Stage 2 (Incipient raceway spall): ringing of bearing/housing natural frequencies (typically 1.5 to 5 kHz) modulated at BPFO, BPFI, or BSF; kurtosis rises above 4.0 and crest factor exceeds 5.0. Action: check lubrication and shorten inspection interval to 2 weeks.
Stage 3 (Discrete fault harmonics in baseband FFT): fundamental BPFO/BPFI/BSF and harmonics (1x, 2x, 3x) with sidebands spaced at shaft speed (BPFI) or FTF (BSF) become visible in the baseband FFT spectrum. Action: schedule bearing replacement within the next planned maintenance window and perform lubricant wear debris analysis.